

Ju Hong Kim

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EXPERIENCE

AMD - GPU Validation Intern

May 2025 - August 2026

- Developed and extended C++ GPU diagnostics to validate compute hardware IP blocks, working directly with GPU command submission, queue management, and register interfaces
- Implemented device-side profiling using AQL completion timestamps by binding completion signaling to kernel dispatch and extracting start/end timestamps for accurate kernel runtime measurement
- Brought up and validated command-processor DMA (CPDMA) engines on pre-silicon emulation platforms, debugging copy/fill flows via register inspection and PM4 packet trace analysis to resolve hangs and misprogrammed sequences
- Designed synchronized multi-die GPU execution using shared ring buffer and doorbell to run single submission across all compute dies, then analysed the drop in current draw during workloads to tune a peak-power stress test
- Ported and validated a sparse-matrix GEMM system-level test ensuring deterministic, bit-accurate results across low-precision data-types (i.e. fp32, fp16, bf16, fp8)

Ericsson - Software Developer Intern

May 2024 - April 2025

- Developed a telemetry pipeline for Linux cold-reboot cycles, collecting CPU, memory, and startup-time metrics to enable system-level visibility and cross-version regression analysis

Blackberry - Systems Software Developer Intern

May 2023 - August 2023

- Built a Protobuf-based proof of concept to evaluate replacing JSON in a C-based telemetry pipeline in preparation for onboarding a larger user base, demonstrating an approximate 50% message size reduction
- Identified a spec-level Avro header encoding bug in Fluent Bit by inspecting binary messages against the Avro specification; documented findings internally and published a technical writeup on the root cause

Blackberry QNX - Student Support Developer

January 2022 - April 2022

- Provided low-level debugging support for applications build on QNX RTOS, diagnosing runtime, memory, and networking issues using GDB, Valgrind, tcpdump, and Wireshark
- Resolved GCC compiler and linker issues including hard/soft float ABI mismatch across RTOS and POSIX C environments

IBM - Db2 Build DevOps Developer

July 2020 - August 2021

- Led early-phase migration of legacy hardware, end-of-life operating systems, and build tooling into secured, isolated environments, adapting scripts and workflows to accommodate evolving security requirements
- Supported build and tooling workflows across AIX, Red Hat, SunOS, and HP-UX, contributing to Clearcase-to-Git transition by identifying RHEL build dependencies

IBM - Db2 Build DevOps Intern

May 2018 - August 2019

- Investigated and triaged nightly build failures in large-scale UNIX/Linux CI builds, distinguishing environmental faults (NFS, permissions, host failures) from genuine C/C++ regressions
- Isolated offending changes across x86, Itanium, PowerPC, and SPARC architectures using targeted rebuilds and log analysis

SKILLS

- **Systems Programming:** C, C++
- **Scripting and other Programming:** Python, Perl, Bash, Java
- **Debugging:** GDB, Valgrind, Wireshark, tcpdump
- **Performance & Profiling:** perf, Callgrind, Flamegraphs
- **Operating System:** Linux, UNIX, QNX (RTOS)
- **Tools:** Git, Make, GCC, Jenkins

PROJECTS

Desktop Passwordless Continuous Authentication System

- Engineered a custom Linux PAM (Pluggable Authentication Module) in C to enable passwordless continuous authentication via Bluetooth device presence over D-Bus.
- Developed a background system daemon using BlueZ (Linux Bluetooth stack) and D-Bus to continuously monitor device proximity, automatically locking sessions on disconnection

16-bit Assembler Implementation

- Developed a two-pass C assembler for a 16-bit ISA, resolving forward references via a symbol table

EDUCATION

Carleton University - HBMath in Mathematics

September 2021 - April 2027

- Teaching Assistant for undergraduate calculus and linear algebra courses

University of Toronto Mississauga - HBSc in Computer Science

September 2015 - June 2020